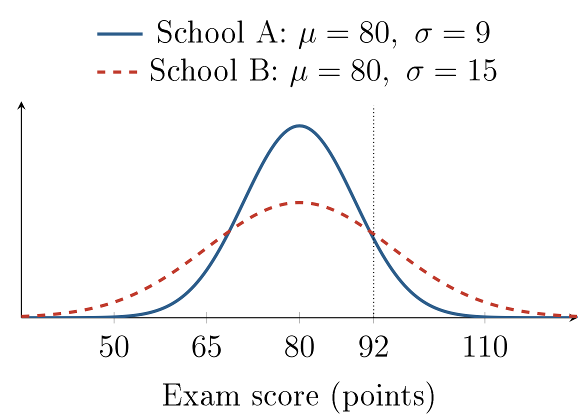
Comparing a Score of 92

Unit 2 · Topic 2.11 · TODAY'S PROBLEM

Suppose exam scores at School A and School B are each approximately normal. Both distributions have mean 80 points, but School A has standard deviation 9 points and School B has standard deviation 15 points.

A counselor says, "A score of 92 is at about the 90th percentile at both schools because both schools have the same mean." A data coach says, "The different spreads could put 92 at different percentiles."



FIRST TAKE (5 min, on your sheet)

Would a 92 stand out equally at both schools? Give one reason from the picture.

DOK 1 **(a)** Identify which school has the more variable score distribution and state how the corresponding normal curve shows that difference.

DOK 2 **(b)** For a score of 92, calculate the z-score and percentile at each school. Show the standardization and report each percentile to the nearest tenth of a percent.

DOK 3 **(c)** In 3–4 sentences, decide whose claim is better supported. Use both percentiles from (b) and explain how the different spreads change what a score of 92 means at each school.

Optional review: follow-along (scan the code)

Work (a)–(c) from the rules on your sheet. Turn in your sheet.